soilice: A modelling framework to test alternative process representations in a frozen soil hydrological model

Andrew Ireson

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1 Developing Governing Equations for soilice

1.1 Internal energy

Consider a control volume of water, of mass M (kg) and temperature T (°C). We define internal energy associated with sensible heat, that is the sensible internal energy, U_s (J), as

$$U_s = M c_p T \tag{1}$$

where c_p (J kg^{-1} K^{-1}) is the specific heat capacity (which is defined by this equation). The differential equation for the change in sensible internal energy with time is

$$\frac{dU_s}{dt} = \frac{d}{dt}(M c_p T) = M c_p \frac{dT}{dt} + c_p T \frac{dM}{dt} \tag{2}$$

Latent heat is a form of internal energy, called latent internal energy, U_l (J), associated with the chemical bonds between molecules in a particular phase (solid, liquid, gas) of a substance. We assume latent heat is negative in the solid phase, zero in the liquid phase and positive in the vapor phase, hence

$$\begin{aligned} U_{li} &= -M_i \lambda_f \\ U_{ll} &= 0 \\ U_{lv} &= -M_v \lambda_v \end{aligned} \tag{3}$$

where λ_f, the amount of heat needed to convert 1 kg of ice to 1 kg of liquid water, is a constant $\approx 3.34 \times 10^5$ J kg^{-1} and $\lambda_v = 2.50 \times 10^6$ J kg^{-1} at $T = 0$°C).

Using this convention, the total latent internal energy is

$$U_l = -M_i \lambda_f + M_v \lambda_v \tag{4}$$

and the change in latent internal energy is

$$\frac{dU_l}{dt} = -\lambda_f \frac{dM_i}{dt} + \lambda_v \frac{dM_v}{dt} \tag{5}$$

The total internal energy that concerns us in soils is given by the sum of sensible and latent heat, i.e.

$$U = U_s + U_l \tag{6}$$

In a control volume containing ice, liquid water, vapor and soil solids we have

$$U = U_{si} + U_{sl} + U_{sv} + U_{ss} + U_{li} + U_{ll} + U_{lv} \tag{7}$$

Adopting our conventions above we can drop U_{ll}.

1.2 Mass and energy balance equations

We express the mass and internal energy per unit volume, i.e., as $\bar{m}$ (kg m^{-3}) and $\bar{u}$ (J m^{-3}). The mass and energy balance equations are

$$\frac{d\bar{m}}{dt} = \frac{d\bar{m}_l}{dt} + \frac{d\bar{m}_i}{dt} = -\nabla q_l \rho_l \tag{8}$$

and

$$\frac{d\bar{u}}{dt} = \frac{d\bar{u}_{si}}{dt} + \frac{d\bar{u}_{sl}}{dt} + \frac{d\bar{u}_{ss}}{dt} + \frac{d\bar{u}_{li}}{dt} = -\nabla(j_a + j_c + j_o) \tag{9}$$

where $\bar{u}_{ss}$ (J m^{-3}) is the sensible internal energy of the soil solids, j_a, j_c and j_o (J m^{-2} s^{-1}) are the advective, conductive, and other heat flux components.

Expanding the left hand side of the energy balance equation gives

$$\frac{d\bar{u}}{dt} = (\bar{m}_i c_{pi} + \bar{m}_l c_{pl} + \bar{m}_s c_{ps})\frac{dT}{dt} + c_{pl}T\frac{d\bar{m}_l}{dt} + (c_{pi}T - \lambda_f)\frac{d\bar{m}_i}{dt} \tag{10}$$

We define the bulk heat capacity, c_{pb} (J kg^{-1} K^{-1}) as

$$c_{pb} = \bar{m}_i c_{pi} + \bar{m}_l c_{pl} + \bar{m}_s c_{ps} \tag{11}$$

so we have

$$\frac{d\bar{u}}{dt} = c_{pb}\frac{dT}{dt} + c_{pl}T\frac{d\bar{m}_l}{dt} + (c_{pi}T - \lambda_f)\frac{d\bar{m}_i}{dt} \tag{12}$$

Now we want to eliminate the terms $d\bar{m}_l/dt$ and $d\bar{m}_i/dt$. Under equilibrium conditions (which is a big assumption) a soil control volume should reach a unique state on the basis of the temperature and the total water. We ignore any hysteresis effects. We define the total water content, θ_T (-) as being given by

$$\theta_T = \theta_L + \frac{\rho_I}{\rho_L}\theta_I \tag{13}$$

We defined ψ_e (m) as the effective matric potential, which is associated with θ_T by the soil characteristic curve relationship,

$$\theta_T = M(\psi_e) \tag{14}$$

ψ_f (m) is the matric potential that corresponds to the freezing threshold, as defined by the GCE, where (Amankwah et al., 2021):

$$\psi_f = \frac{\lambda_f}{g}\ln\left(\frac{T + T_0}{T_0}\right) \tag{15}$$

where T is the soil temperature in °C, and $T_0 = 273.15$. If $\psi_e \leq \psi_f$ there is no ice in the soil pore space. Now we say

$$\begin{aligned} &\theta_T = M(\psi_e); \quad \theta_L = \theta_T; && \theta_I = 0; && \text{if } \psi_e \leq \psi_f \\ &\theta_T = M(\psi_e); \quad \theta_L = M(\psi_f(T)) = F(T); && \theta_I = \tfrac{\rho_L}{\rho_I}(\theta_T - \theta_L); && \text{if } \psi_e > \psi_f \end{aligned} \tag{16}$$

The function $F(T)$ defines the freezing characteristic curve, and is simply the combination of the GCE and the $\theta = M(\psi)$ relationship. $M(\psi)$ and $F(T)$ can be differentiated, giving

$$\frac{d\theta_T}{d\psi_e} = M'(\psi_e) \tag{17}$$

and

$$\frac{d\theta_L}{dT} = G'(T) \tag{18}$$

So we can further say

$$\begin{aligned}
&\frac{d\theta_T}{dt} = M'(\psi_e)\frac{d\psi_e}{dt}; \quad \frac{d\theta_L}{dt} = \frac{d\theta_T}{dt}; \qquad \frac{d\theta_I}{dt} = 0; \qquad \text{if} \;\; \psi_e \leq \psi_f \\
&\frac{d\theta_T}{dt} = M'(\psi_e)\frac{d\psi_e}{dt}; \quad \frac{d\theta_L}{dt} = F'(T)\frac{dT}{dt}; \quad \frac{d\theta_I}{dt} = \frac{\rho_L}{\rho_I}\left(\frac{d\theta_T}{dt} - \frac{d\theta_L}{dt}\right); \quad \text{if} \;\; \psi_e > \psi_f
\end{aligned} \tag{19}$$

Now, consider first the case where $\psi_e \leq \psi_f$ (i.e. the soil is unfrozen). We have

$$\frac{d\bar{m}_l}{dt} = \frac{d\bar{m}}{dt}; \quad \frac{d\bar{m}_i}{dt} = 0 \tag{20}$$

Substituting this into Equation 12 we have

$$\frac{dT}{dt} = \frac{\frac{d\bar{u}}{dt} - c_{pl}T\frac{d\bar{m}}{dt}}{c_{pb}} \tag{21}$$

Next consider the case where $\psi_e > \psi_f$ (i.e. the soil is frozen). This time we have

$$\frac{d\bar{m}_l}{dt} = \rho_l F'(T)\frac{dT}{dt} \tag{22}$$

and

$$\frac{d\bar{m}_i}{dt} = \frac{d\bar{m}}{dt} - \frac{d\bar{m}_l}{dt} = \frac{d\bar{m}}{dt} - \rho_l F'(\bar{m}, T)\frac{dT}{dt} \tag{23}$$

Substituting these expressions into the Equation 12 we get

$$\frac{d\bar{u}}{dt} = (c_{pb} + \rho_l F'(T)((c_{pl} - c_{pi})T + \lambda_f))\frac{dT}{dt} + (c_{pi}T - \lambda_f)\frac{d\bar{m}}{dt} \tag{24}$$

Rearranging we have

$$\frac{dT}{dt} = \frac{\frac{d\bar{u}}{dt} - (c_{pi}T - \lambda_f)\frac{d\bar{m}}{dt}}{c_{pb} + \rho_l F'(T)((c_{pl} - c_{pi})T + \lambda_f)} \tag{25}$$

which may then be solved by replacing the derivative terms on the right hand side with the net fluxes.

1.3 Finite difference formulation

For a one-dimensional vertical model, with z (m) representing the depth below the ground surface, it is good practice to use a block-centred grid, where z_n denotes the midpoint of a cell, and $z_{n-1/2}$ and $z_{n+1/2}$ are the depths of the top and bottom faces of the cell, respectively. Considering a single control volume at depth index n, we can write

$$\begin{aligned} \left.\frac{d\bar{m}_l}{dt}\right|_n + \left.\frac{d\bar{m}_i}{dt}\right|_n &= -\rho_l \nabla q_n \\ \left.\frac{d\bar{u}}{dt}\right|_n &= -\nabla j_n \end{aligned} \tag{26}$$

where ∇q_n and ∇j_n are the net fluxes of mass and energy into the control volume. Expressing these equations with total mass, $\bar{m}$ and temperature T as the dependent variables we have

$$\left.\frac{d\bar{m}}{dt}\right|_n = -\rho_l \nabla q_n \tag{27}$$

$$\left.\frac{dT}{dt}\right|_n = \begin{cases} \dfrac{\nabla j_n - c_{pl} T_n \left.\frac{d\bar{m}}{dt}\right|_n}{c_{pb}}, & \psi_{e,n} \leq \psi_{f,n} \\ \dfrac{\nabla j_n - (c_{pi} T_n - \lambda_f) \left.\frac{d\bar{m}}{dt}\right|_n}{c_{pb} + \rho_l f'(\bar{m}_n, T_n)((c_{pl} - c_{pi})T_n + \lambda_f)}, & \psi_{e,n} > \psi_{f,n} \end{cases} \tag{28}$$

Expanding the flux terms we have

$$\begin{aligned} -\rho_l \nabla q_n &= \frac{(q_{l,n-1/2} - q_{l,n+1/2})\rho_l}{z_{n+1/2} - z_{n-1/2}} \\ -\nabla j_n &= \frac{(j_{a,n-1/2} - j_{a,n+1/2}) + (j_{c,n-1/2} - j_{c,n+1/2}) + (j_{o,n-1/2} - j_{o,n+1/2})}{z_{n+1/2} - z_{n-1/2}} \end{aligned} \tag{29}$$

Here the individual flux terms are given by

$$\begin{aligned} q_{l,n-1/2} &= -K_{n-1/2}\left(\tfrac{\psi_n - \psi_{n-1}}{z_i - z_{n-1}} - 1\right) \\ q_{l,n+1/2} &= -K_{n+1/2}\left(\tfrac{\psi_{n+1} - \psi_n}{z_{n+1} - z_n} - 1\right) \\ j_{a,n-1/2} &= q_{l,n-1/2}\rho_l c_{pl} T_{n-1} \\ j_{a,n+1/2} &= q_{l,n+1/2}\rho_l c_{pl} T_n \\ j_{c,n-1/2} &= -\kappa_{n-1/2}\left(\tfrac{T_n - T_{n-1}}{z_n - z_{n-1}}\right) \\ j_{c,n+1/2} &= -\kappa_{n+1/2}\left(\tfrac{T_{n+1} - T_n}{z_{n+1} - z_n}\right) \end{aligned} \tag{30}$$

For the advected heat fluxes here we assume the water flux is positive in the z direction, such that T_{n-1} is the downstream temperature at point $n - 1/2$, and T_n is the upstream temperature at point $n + 1/2$. Alternative formulations for advection may be considered, but we will not expand on this here. The term j_o will be zero for all n except $n = 1$, that is the upper boundary condition (i.e. $j_{o,1/2} \neq 0$).

Consider a soil profile subject to a vertical infiltration flux q_a (m s^{-1}), a drainage flux, q_d (m s^{-1}) and an evaporative loss term q_e (m s^{-1}). The following boundary conditions can be used, where index $n = 1$ represents the ground surface and $n = N$ is the lower boundary:

$$
\begin{aligned}
q_{l,1/2} &= q_a - q_e \\
q_{l,N+1/2} &= q_d \\
j_{a,1/2} &= j_{a,T} \\
j_{c,1/2} &= j_{g,T} \\
j_{a,N+1/2} &= j_{a,B} \\
j_{c,N+1/2} &= j_{g,B} \\
j_{o,1/2} &= j_r - j_h - j_e \\
j_{o,N+1/2} &= 0
\end{aligned}
\tag{31}
$$

where all j terms have units (J m^{-2} s^{-1}), and $j_{a,T}$ is advection with net infiltrating water, $j_{g,T}$ is the ground heat flux, driven by conduction across the soil surface (important when there is snow on the ground), j_r is net radiation, j_h is the sensible heat lost (advection with the movement of air), j_e is the latent heat lost to evaporation (advection of water vapor), $j_{a,B}$ is advection with water draining out the base of the soil and $j_{g,B}$ is the conductive heat flux across the lower boundary. The different flux terms in Eq 31 may be zero under different scenarios (e.g. snow cover, bare soil, etc) or assumptions.

We have here a system of ordinary differential equations which we solve using an ODE solver with the method of lines.

1.4 Constituitive relationships

Now, we need expressions for M, M', F and F', all of which we can get from the van Genuchten equations combined with the GCE equation.

$$S_e = (1 + (\alpha\psi)^n)^{-m} \tag{32}$$

$$\theta = M(\psi) = \theta_r + (\theta_s - \theta_r)S_e \tag{33}$$

$$\frac{d\theta}{d\psi} = M'(\psi) = \frac{-\alpha m(\theta_s - \theta_r)}{1-m} S_e^{1/m} \left(1 - S_e^{1/m}\right)^m \tag{34}$$

The GCE:

$$\psi_f(T) = \frac{\lambda_f}{g} \ln\left(\frac{T + T_0}{T_0}\right) \approx \frac{L_f}{g}\left(\frac{T}{T_0}\right) \tag{35}$$

and

$$
\begin{aligned}
F(T) &= M\left(\psi_f(T)\right) \\
&= \theta_r + (\theta_s - \theta_r)\left(1 + \left(\tfrac{\alpha L_f T}{g T_0}\right)^n\right)^{-m}
\end{aligned}
\tag{36}
$$

For the derivative of the SFC let us define

$$S_f = \left(1 + \left(\frac{\alpha L_f T}{g T_0}\right)^n\right)^{-m} \tag{37}$$

Then

$$F'(T) = \frac{-\alpha L_f m(\theta_s - \theta_r)}{g T_0 (1-m)} S_f^{1/m}\left(1 - S_f^{1/m}\right)^m \tag{38}$$

Finally we need to define the hydraulic conductivity. In unfrozen conditions this is

$$K(\psi) = K_s S_e^{1/2} \left(1 - \left(1 - S_e^{1/m}\right)^m\right) \tag{39}$$

In unfrozen conditions, we are still seeking an ideal relationship for $K(\psi_e, T)$. For now we use impedence.